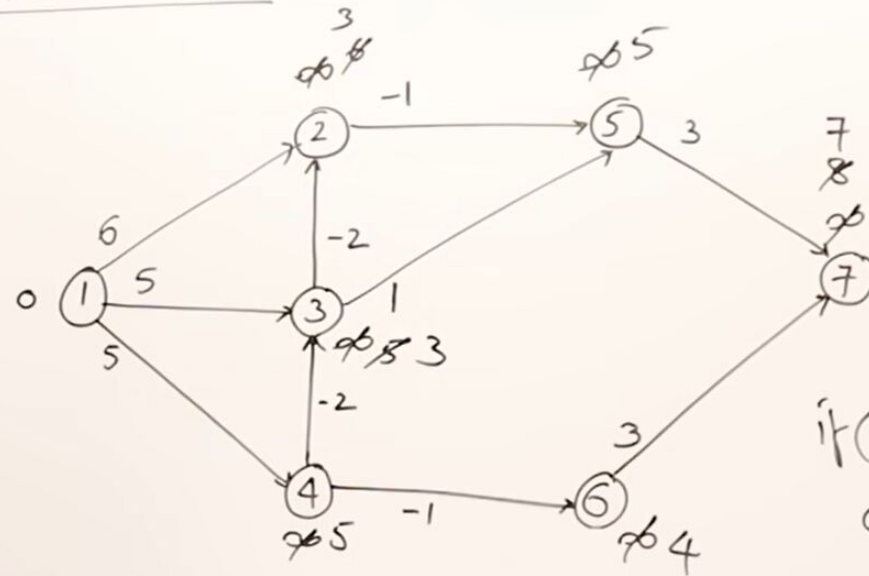


Single Source Shortest Path Bellman-Ford



(u, v)

Relaxation

if $(d[u] + c(u, v) < d[v])$
 $d[v] = d[u] + c(u, v)$

EdgeList $\rightarrow (1, 2), (1, 3), (1, 4), (2, 5), (3, 2), (3, 5), (4, 3), (4, 6), (5, 7), (6, 7)$

this is less so change so most of the changes we have seen first time only



Single Source Shortest Path

Bellman-Ford

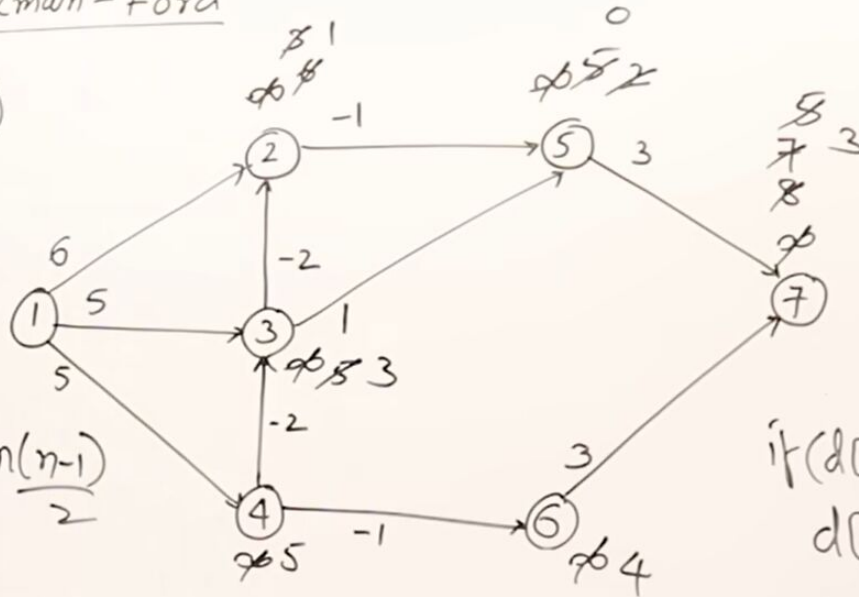
1-0
2-1
3-3
4-5
5-0
6-4
7-3

$$O(|E||V|-1)$$

$$O(|V||E|)$$

$$O(n^2)$$

$$|E| = \frac{n(n-1)}{2}$$



(u,v)

Relaxation

if $(d[u] + c(u,v) < d[v])$
 $d[v] = d[u] + c(u,v)$

edgeList $\rightarrow (1,2), (1,3), (1,4), (2,5), (3,2), (3,5), (4,3), (4,6), (5,7), (6,7)$
 $\frac{n(n-1)}{2} \times (n-1) = O(n^3)$

this one this will be order of n cube so
in case



Single Source Shortest Path

Bellman-Ford

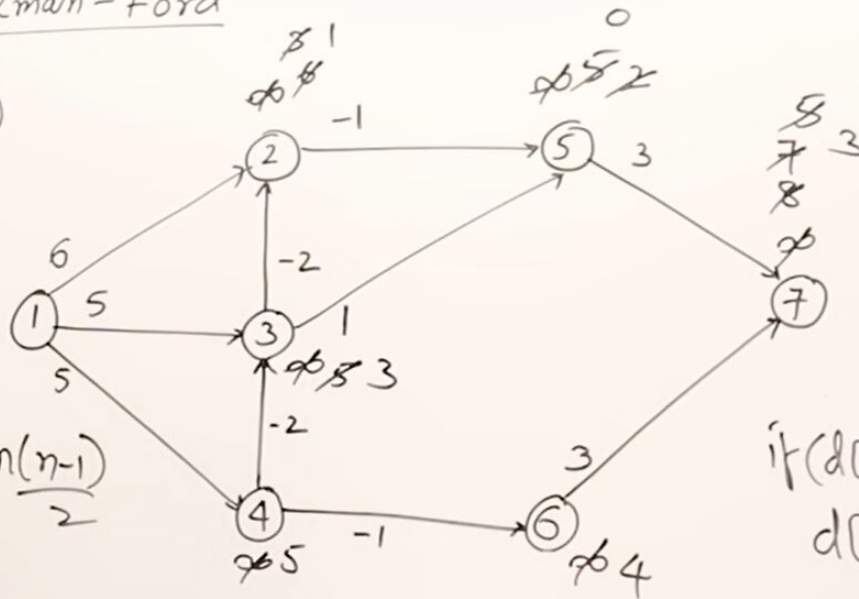
$$O(|E||V|-1)$$

$$O(|V||E|)$$

$$O(n^2)$$

$$|E| = \frac{n(n-1)}{2}$$

$$\frac{n(n-1)}{2} \times (n-1) = O(n^3)$$



(u, v)

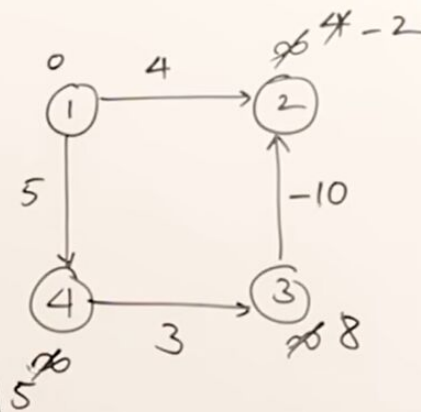
Relaxation

$$\text{if } (d[u] + c(u, v) < d[v]) \\ d[v] = d[u] + c(u, v)$$

time we say it takes n square and at most it may take any cubed time that



Single Source Shortest Path Bellman-Ford



1 - 0
2 - (-2)
3 - 8
4 - 5

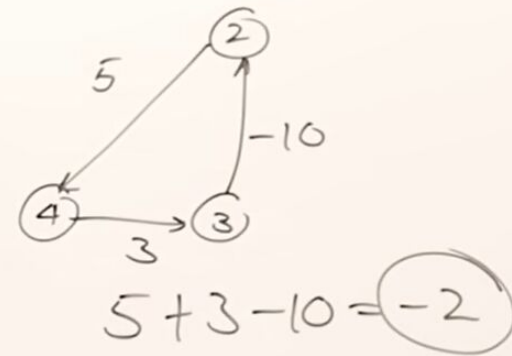
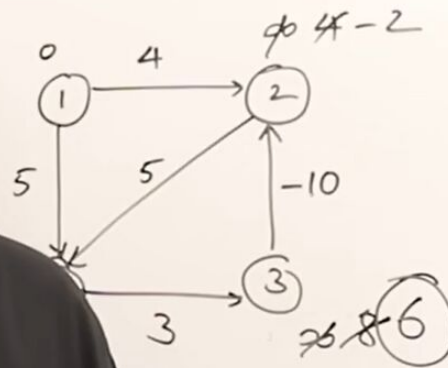
edges → (3,2) (4,3) (1,4) (1,2)

$$n = 4 - 1 = 3$$

extra also there is no change
that means the answers are perfect



Single Source Shortest Path Bellman-Ford

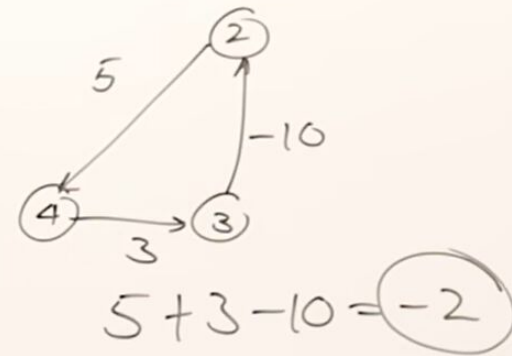
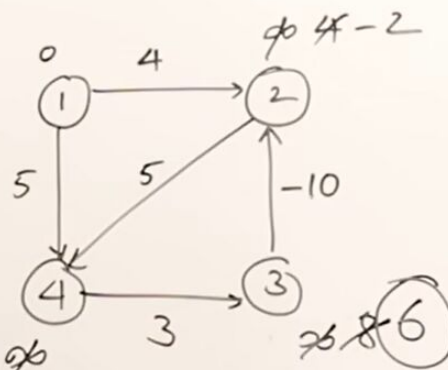


edges $\rightarrow (3,2) (4,3) (1,4) (1,2) (2,4)$

there is a negative weight a cycle then
graph cannot be solved



Single Source Shortest Path Bellman-Ford



edges $\rightarrow (3,2) (4,3) (1,4) (1,2) (2,4)$

cycle total weight of a cycle is
negative if it is positive it will work

